中國海洋大學海德学院

HAIDE COLLEGE, OCEAN UNIVERSITY OF CHINA

Statistical Modelling and Inference

Weeks 13&14

Outline:

Maximum Likelihood Estimation

Fisher Information

Poisson distribution

Suppose $y_1, y_2, \ldots, y_n$ independent and identically distributed from Poisson distribution, $P(\lambda)$.

Find the maximum likelihood estimate of λ.

$$\boldsymbol{f(y_i) = \frac{\lambda^{y_i} e^{-\lambda}}{y_i!}}$$

where $y_i = 0,1,2,3, \ldots, \infty$ and $0 \leq \lambda \leq \infty$

Binomial distribution

Suppose $y_1, y_2, \dots, y_n$ independent and identically distributed from binomial distribution, $Binomial(n, p)$.

Find the maximum likelihood estimate of p.

$$\boldsymbol{f(y_i) = \binom{n}{y_i} p^{y_i} (1 - p)^{n - y_i}}$$

where $y_i = 0,1,2,3, \dots, \infty$ and $0 \le p \le 1$

Exponential distribution

Suppose $y_1, y_2, \dots, y_n$ independent and identically distributed from exponential distribution, $Exp(\theta)$.

Find the maximum likelihood estimate of θ.

$$\boldsymbol{f(y_i) = \theta e^{-\theta y_i}}$$

where $y_i = 0,1,2,3, \dots, \infty$ and $\theta > 0$

Normal distribution (unknown mean $\boldsymbol{\mu}$)

Suppose $y_1, y_2, \dots, y_n$ independent and identically distributed from normal distribution $N(\mu, \sigma^2)$, with **unknown mean $\boldsymbol{\mu}$** and known variance σ^2.

Find the maximum likelihood estimate of μ.

$$\boldsymbol{f(y_i) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2}(y_i-\mu)^2}}$$

where $y_i = 0,1,2,3, \dots, \infty$ with $-\infty \leq \mu \leq +\infty$ and $\sigma^2 > 0$

Normal distribution (unknown variance σ^2)

Suppose $y_1, y_2, \dots, y_n$ independent and identically distributed from normal distribution $N(\mu, \sigma^2)$, with known mean μ and **unknown variance $\boldsymbol{\sigma^2}$**.

Find the maximum likelihood estimate of σ^2.

$$\boldsymbol{f(y_i) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2}(y_i-\mu)^2}}$$

where $y_i = 0,1,2,3, \dots, \infty$ with $-\infty \leq \mu \leq +\infty$ and $\sigma^2 > 0$

Normal distribution (unknown μ and σ^2)

Suppose $y_1, y_2, \dots, y_n$ independent and identically distributed from normal distribution $N(\mu, \sigma^2)$, with **unknown mean $\boldsymbol{\mu}$ and variance $\boldsymbol{\sigma^2}$**.

Find the maximum likelihood estimates of μ and variance σ^2 respectively.

$$\boldsymbol{f(y_i) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2}(y_i-\mu)^2}}$$

where $y_i = 0,1,2,3,\dots,\infty$ with $-\infty \le \mu \le +\infty$ and $\sigma^2 > 0$

Fisher Information

Suppose $y_1, y_2, \dots, y_n$ independent and identically distributed from any distribution with unknown parameter (θ), then the fisher information is:

$$\boldsymbol{I(\theta) = -E\left[\frac{d^2}{d\theta^2} lnL(\underline{y}_i, \theta)\right]}$$

Find the Fisher Information for

1. Poisson distribution
2. Exponential distribution
3. Normal distribution (unknown μ)
4. Normal distribution (unknown σ^2)
5. Normal distribution (unknown μ and σ^2)

Thank

You

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